3.(i) 7 (ii) 9 (iii) $\frac{1}{27}$ (iv) $\frac{25}{16}$

4.(i) $5^{\frac{1}{2}}$ (ii) $7^{\frac{1}{2}}$ (iii) $7^{\frac{10}{3}}$ (iv) $10^{\frac{-14}{3}}$

5.(i) 2 (ii) 3 (iii) 10 (iv) $\frac{4}{5}$

Exercise 2.6

1.(i) $21\sqrt{3}$ (ii) $3\sqrt[3]{5}$ (iii) $26\sqrt{3}$ (iv) $8\sqrt[3]{5}$

2. (i) $\sqrt{30}$ (ii) $\sqrt{5}$ (iii) 30 (iv) $49a - 25b$

(v) $\frac{5}{16}$ 3.(i) 1.852 (ii) 23.978

4. (i) $\sqrt[3]{5} > \sqrt[6]{3} > \sqrt[9]{4}$ (ii) $\sqrt{\sqrt{3}} > \sqrt[2]{\sqrt[3]{5}} > \sqrt[3]{\sqrt[4]{7}}$

5. (i) yes (ii) yes (iii) yes (iv) yes

6. (i) yes (ii) yes (iii) yes (iv) yes

Exercise 2.7

1.(i) $\frac{\sqrt{2}}{10}$ (ii) $\frac{\sqrt{5}}{3}$ (iii) $\frac{5\sqrt{6}}{6}$ (iv) $\frac{\sqrt{30}}{2}$

2. (i) $\frac{4}{3}(5 + 2\sqrt{6})$ (ii) $13 - 4\sqrt{6}$ (iii) $\frac{9 + 4\sqrt{30}}{21}$ (iv) $-2\sqrt{5}$

3. $a = \frac{-4}{3}, b = \frac{11}{3}$ 4. $x^2 + \frac{1}{x^2} = 18$ 5. 5.414

Exercise 2.8

1. (i) 5.6943×10^{11} (ii) 2.00057×10^{3} (iii) 6.0×10^{-7} (iv) 9.000002×10^{-4}

2. (i) 3459000 (ii) 56780 (iii) 0.0000100005 (iv) 0.0000002530009

3. (i) 1.44×10^{28} (ii) 8.0×10^{-60} (iii) 2.5×10^{-36}

4.(i) 7.0×10^{9} (ii) 9.4605284×10^{15} km (iii) $9.1093822 \times 10^{-31}$ kg

5. (i) 1.505×10^{8} (ii) 1.5522×10^{17} (iii) 1.224×10^{7} (iv) 1.9558×10^{-1}

Exercise 2.9

1. (4) 2. (3) 3. (2) 4. (1) 5. (4) 6. (2) 7. (2) 8. (2) 9. (4) 10. (1)

11. (4) 12. (4) 13. (4) 14. (2) 15. (2) 16. (3) 17. (2) 18. (4) 19. (2) 20. (3)

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